Module 3 Final Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 99,990 of 99,990 points. 100%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though — spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

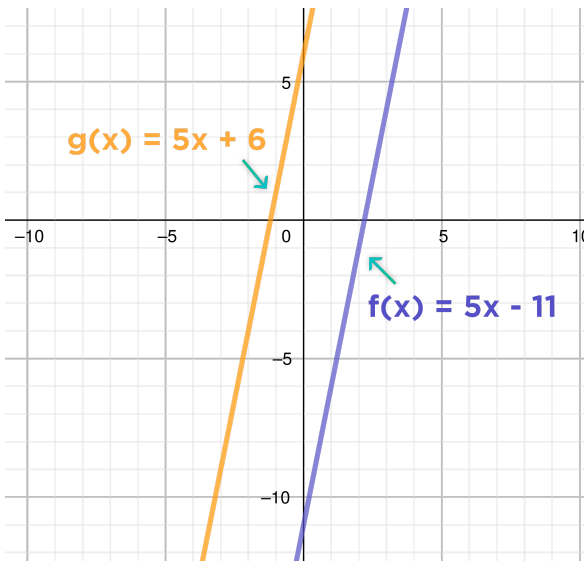
1 Find the vertical distance between two parallel lines given by the following equations:

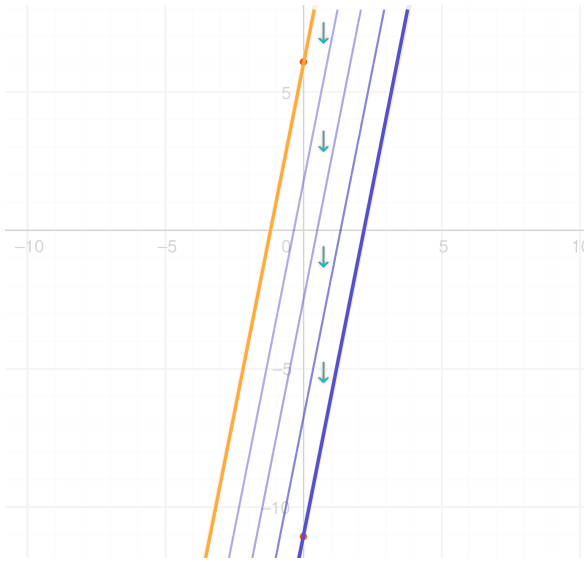
$$y(x) = 5x - 11$$
$$g(x) = 5x + 6$$

Your Answer	17
Correct Answer	17

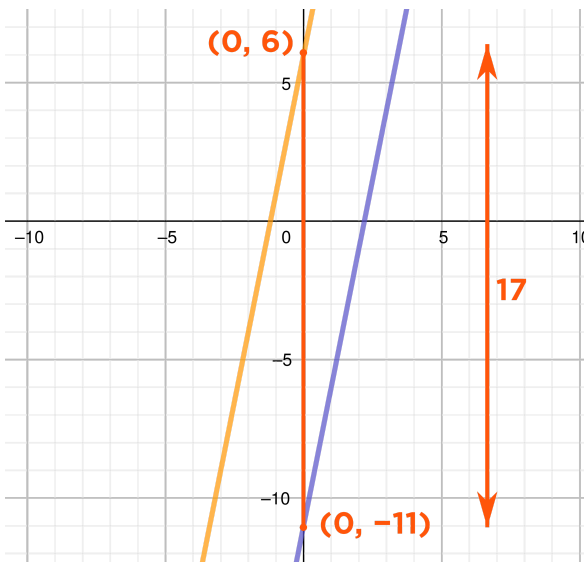
Explanation

Since these two lines are parallel, you can think of one equation as being a shifted-up version of the other line. The distance between these two lines is equal to this vertical shift. Look at the y-axis to see the amount of this shift:





$$y(x) \text{ crosses the } y\text{-axis at } x = 0 \implies y(0) = 5(0) - 11 = -11$$
$$g(x) \text{ crosses the } y\text{-axis at } x = 0 \implies g(0) = 5(0) + 6 = 6$$



The difference between these y-intercepts is $6 + 11 = \boxed{17}$, so this is the answer.

2 There are 100 cycles, all of which are bicycles (which have 2 wheels) or tricycles (which have 3) wheels. The cycles have 237 wheels altogether. How many bicycles are there?

✓	**Your Answer**	63
	Correct Answer	63

Explanation

Let $B =$ the number of bicycles and $T =$ the number of tricycles. Then $2B + 3T = 237$ and $B + T = 100$. Substitute $100 - T$ for B to eliminate B, and we get $2\left(100 - T\right) + 3T = 237$, which simplifies to $200 - 2T + 3T = 237 \implies T = 37 \implies B = 100 - 37 = \boxed{63}$.

Alternatively, suppose that all 100 were bicycles. Then there would be $100 \times 2 = 200$ wheels. Switching a bicycle into a tricycle gains 1 wheel. We need to gain $237 - 200 = 37$ wheels, so we should have 37 tricycles and therefore $\boxed{63}$ bicycles.

3

Suppose that the symbol $x \mathbin{@} y$ means $\frac{1}{x} + \frac{1}{y}$. If $3 \mathbin{@} (2 \mathbin{@} x) = 1$, what is x?

Your Answer	1	
Correct Answer	1	

Explanation

We must have

$$\begin{aligned} \frac{1}{3} + \frac{1}{2 \mathbin{@} x} &= 1 \\ \frac{1}{2 \mathbin{@} x} &= \frac{2}{3} \\ (2 \mathbin{@} x) &= \frac{3}{2} \\ \frac{1}{2} + \frac{1}{x} &= \frac{3}{2} \\ \frac{1}{x} &= 1 \\ x &= \boxed{1} \end{aligned}$$

4 There are three 2-digit numbers that have the property that if you subtract 3 times its tens digit from it, the result is the original 2-digit number with the digits swapped.

Find the sum of these three 2-digit numbers.

✓	**Your Answer**	192
	Correct Answer	192

Explanation

Let the original number be $10T + U$. Then, the condition is equivalent to:

$$\begin{aligned}(10T + U) - 3T &= 10U + T \\ 6T &= 9U.\end{aligned}$$

The three choices that work for T and U are single digit numbers in the ratio of $3:2$. The possibilties are:

$$\begin{aligned}&T = 9 \text{ and } U = 6 \\ &T = 6 \text{ and } U = 4, \text{ and} \\ &T = 3 \text{ and } U = 2.\end{aligned}$$

Each number is equal to $10T + U$, so the three numbers are $96, 64$ and 32, and their sum is $96 + 64 + 32 = \boxed{192}$.

5 What is the sum of all even 2-digit numbers?

Your Answer	2430
Correct Answer	2430

Explanation

The number of 2-digit numbers is 90, and half of them are even. They form an arithmetic progression with first term 10 and last term 98.

$$\begin{aligned}\text{Sum } &= 10+12+14+16+18+20+22+\ldots+94+96+98\\ &= 2\times 5+2\times 6+2\times 7+2\times 8+2\times 9+2\times 10+2\times 11+\ldots+2\times 47+2\times 48+2\times 49\end{aligned}$$

The number of numbers is $49-5+1=45$. Their average is $\frac{10+98}{2}=\frac{108}{2}=54$. So, their sum is $45\times 54=\boxed{2430}$.

6 A square's area plus its perimeter equals 285. What is the side length of the square?

Your Answer	15
Correct Answer	15

Explanation

Let's let x be the side length of the square; then the area is x^2 and the perimeter is $4x$. We can write

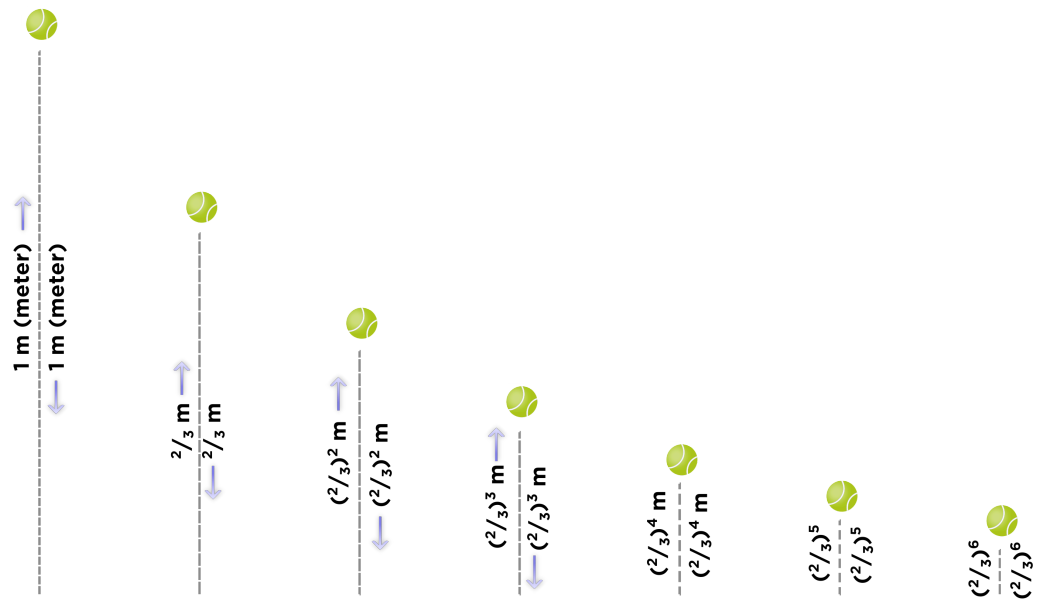
$$x^2 + 4x = 285 \implies x^2 + 4x - 285 = 0.$$

Factoring, we get $(x + 19)(x - 15) = 0$. Since the side length must be positive, the only logical choice is $x = \boxed{15}$.

7 Starting from a point on the ground, a ball is thrown straight up to a height of 1 meter. It bounces up and down vertically, each time reaching a maximum height that is $\frac{2}{3}$ its previous maximum height. How many total meters will the ball travel, if it keeps bouncing forever?

✓	**Your Answer**	6
	Correct Answer	6

Explanation



We can write

$$2 \cdot \left(1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \ldots\right) = x,$$

where is x is the total distance the ball travels. Then $x - 2 = \frac{2}{3}x \implies x = \boxed{6}$.

8 Simplify $\sqrt{20+10\sqrt{3}}$ and rewrite it in simplest radical form as $\sqrt{a}\left(b+\sqrt{c}\right)$ by filling whole numbers into these blanks:

√ 1. × (2. + √ 3.)

Your Answer

1. 5
2. 1
3. 3

Correct Answer

1. 5
2. 1
3. 3

Explanation

We ultimately want to find a way for $(\sqrt{x}+\sqrt{y})^2 = 20+10\sqrt{3} = 20+2\sqrt{75}$. Expanding, this would be $x+2\sqrt{xy}+y$, so by matching coefficients we find that we need:

$$\begin{aligned} x+y &= 20 \\ xy &= 75. \end{aligned}$$

By inspection, this happens when choosing $x=15$ and $y=5$, or vice versa. The order does not matter because we are adding their square roots. The answer then simplifies to:

$$\begin{aligned} \sqrt{15}+\sqrt{5} &= \sqrt{3}\sqrt{5}+\sqrt{5} \\ &= \boxed{\sqrt{5}\left(1+\sqrt{3}\right)} \end{aligned}$$

9 If Brian bikes to school at a speed of 15 miles per hour, he will get to school two minutes early. If he bikes to school at a speed of 12 miles per hour, he will get to school two minutes late. What speed should he bike at so that he gets to school on time?

Your answer should be an improper fraction in simplest terms without spaces; e.g., if the answer is 13/2, you would put 13 in the first blank and 2 in the second blank.

_1._ / _2._

Your Answer

1. 40
2. 3

Correct Answer

1. 40
2. 3

Explanation

Converting their speeds from hours into minutes, we get

$$\begin{aligned}\text{Brian's faster speed } &= 15\frac{\text{miles}}{\text{hour}} = \frac{15}{60}\frac{\text{miles}}{\text{min}} = \frac{1}{4}\frac{\text{miles}}{\text{min}}\\ \text{Brian's slower speed } &= 12\frac{\text{miles}}{\text{hour}} = \frac{12}{60}\frac{\text{miles}}{\text{min}} = \frac{1}{5}\frac{\text{miles}}{\text{min}}\end{aligned}$$

Let's let $t = \text{ time it takes if Brian is on time.}$ Using the distance equation, we know

$$\begin{aligned}\text{distance } &= \text{ speed } \times \text{ time}\\ &= \frac{1}{4}\frac{\text{miles}}{\text{min}}(t-2)\ (\text{Faster speed})\\ &= \frac{1}{5}\frac{\text{miles}}{\text{min}}(t+2)\ (\text{Slower speed})\end{aligned}$$

Setting these equal to each other, we can solve for t :

$$\begin{aligned}\frac{1}{4}(t-2) &= \frac{1}{5}(t+2)\\ \frac{1}{4}t - \frac{1}{2} &= \frac{1}{5}t + \frac{2}{5}\\ \frac{1}{20}t &= \frac{2}{5} + \frac{1}{2}\\ \frac{1}{20}t &= \frac{9}{10}\\ t &= \frac{9}{10} \times 20\\ &= 18 \text{ minutes}\end{aligned}$$

Plugging this back into the equation for distance, we can solve for v, the speed Brian needs to go at in order to reach school in time.

$$\begin{aligned}
\text{distance } &= \frac{1}{4}(18-2) = 18v \\
\implies 4 &= 18v \\
\implies v &= \frac{4}{18} \\
&= \frac{2}{9}\frac{\text{miles}}{\text{min}}
\end{aligned}$$

This is in miles per minute, so we can convert to miles per hour

$$\begin{aligned}
v &= \frac{2}{9}\frac{\text{miles}}{\text{min}} \times 60\frac{\text{min}}{\text{hour}} \\
&= \frac{2}{3} \times 20\frac{\text{miles}}{\text{hour}} \\
&= \boxed{\frac{40}{3}\text{ mph}}
\end{aligned}$$

10 Find the value of the positive number:

$$\sqrt{30+\sqrt{30+\sqrt{30+\cdots}}}$$

✓	**Your Answer**	6
	Correct Answer	6

Explanation

Let x denote the answer. We can square both sides to get $x^2 - 30 = x \implies x^2 - x - 30 = 0 \implies (x-6)(x+5) = 0 \implies x = 6 \text{ or } x = -5$. Since we want a positive value of x, $x = \boxed{6}$.

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 The sum of two numbers is 16 and the sum of their cubes is 1456. What is the product of the two numbers?

Your Answer	[no answer given]
Correct Answer	55

Explanation

Cubing $x+y$, we get the expression

$$\begin{aligned}(x+y)^3 &= x^3+3x^2y+3xy^2+y^3\\ &= x^3+3xy(x+y)+y^3\\ &= \left(x^3+y^3\right)+3\,(xy)(x+y)\end{aligned}$$

Plugging in our values of $x^3+y^3=1456$ and $x+y=16$, we get

$$\begin{aligned}1456+3xy\times 16 &= 16^3\\ xy &= \frac{16^3-1456}{48}\\ &= \frac{16^2-91}{3}\\ &= \frac{256-91}{3}\\ &= \frac{165}{3}\\ &= \boxed{55}\end{aligned}$$

12 Two numbers have a difference of 4 with an arithmetic mean 7 larger than their geometric mean.
Find their sum.

Your answer should be a fraction in simplest terms, e.g. if the answer were 1/2, you would put 1 in the first blank, and 2 in the second blank.

_1._ / _2._

Your Answer

1. [no answer given]
2. [no answer given]

Correct Answer

1. 53
2. 7

Explanation

$$\begin{aligned}
\frac{x+y}{2} &= \sqrt{xy}+7 \\
x+y &= 2\sqrt{xy}+14 \\
x+y-14 &= 2\sqrt{xy} \\
(x+y-14)^2 &= (2\sqrt{xy})^2 \\
x^2+xy-14x+xy+y^2-14y-14x-14y+4\times 7^2 &= 4xy \\
x^2-2xy+y^2-28x-28y+4\times 7^2 &= 0 \\
(x-y)^2-28\,(x+y)+4\times 7^2 &= 0
\end{aligned}$$

$$28\,(x+y) = (x-y)^2+4\times 7^2$$

$$x+y = \frac{(x-y)^2+4\times 7^2}{28}$$

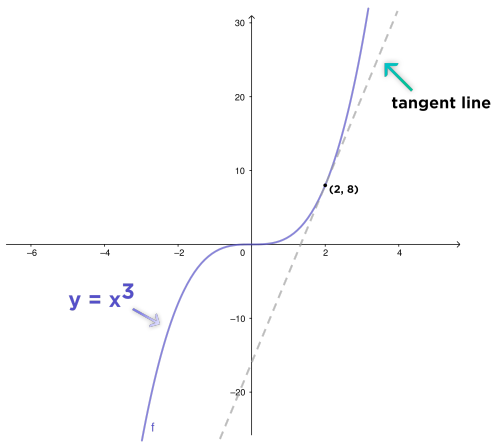
$$\begin{aligned}
x+y &= \frac{4^2}{28}+\frac{4\times 7^2}{28} \\
&= \frac{4}{7}+7 \\
&= \frac{4}{7}+\frac{49}{7} \\
&= \boxed{\frac{53}{7}}
\end{aligned}$$

13 What is the slope of the curve $y = x^3$ at the point $(2, 8)$?

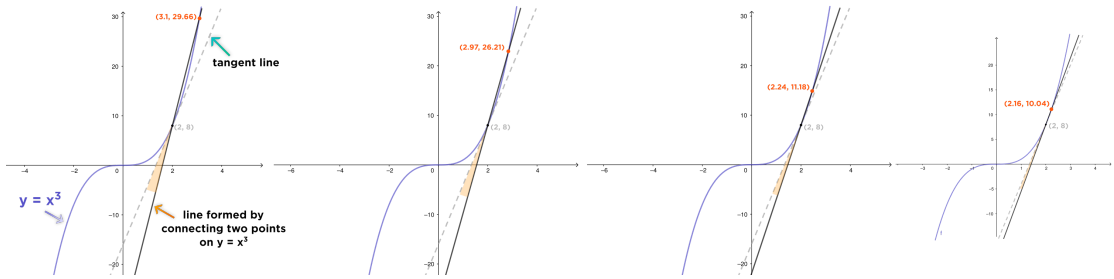
Your Answer	[no answer given]
Correct Answer	12

Explanation

The graph of $y = x^2$ is shown below, with the tangent line at $(2, 8)$ drawn as a dotted line.



If we use a magnifying glass to zoom in on the point $(2, 8)$, and draw a line going through $(2, 8)$ and another point on the $y = x^3$ curve very close to $(2, 8)$, we see that the slope of this line is very close to the slope of the tangent linne at $(2, 8)$.



Let's mark off two points on this tangent line: one, $(2, 8)$, and the other, found by increasing 2 by a tiny bit (we can call this ϵ, a standard variable that represents an infinitesimal tiny bit. Then this other point has an x-value of $2 + \epsilon$ and a y-value of $(2 + \epsilon)^3$. The slope of this line is equal to

$$\frac{\text{change in y}}{\text{change in x}} = \frac{(2+\epsilon)^3 - 2^3}{\epsilon}$$

Expanding, we get

$$\begin{aligned}
&\frac{\left(4+4\epsilon+\epsilon^2\right)(2+\epsilon)-2^3}{\epsilon} \\
&= \frac{8+4\epsilon+8\epsilon+4\epsilon^2+2\epsilon^2+\epsilon^3-8}{\epsilon} \\
&= \frac{12\epsilon+6\epsilon^2+\epsilon^3}{\epsilon} \\
&= 12+6\epsilon+\epsilon^2
\end{aligned}$$

The trick here is that as $\epsilon \to 0$, the ϵ terms vanish, so the slope reduces to $\boxed{12}$.

14 Two friends, R.U. Cumming and Ida Ngo, live 7 miles apart. R. U. starts to walk toward Ida's house, texting her along the way. After receiving R.U.'s text, Ida decides to walk towards his house, at $\frac{6}{5}$ his speed. They meet somewhere in between, after R.U. has spent 3 times as long walking as Ida. How many miles did R. U. walk?

Your answer should be the number only, without units.

Your Answer	[no answer given]
Correct Answer	5

Explanation

If R.U.'s speed were equal to w, then Ida's speed would be $\frac{6}{5}w$. In other words, if R.U.'s speed were $5v$, then Ida's speed would be $6v$. So for every 5 miles that R.U. goes, Ida can go 6 miles. R.U. spends three times as much time walking as Ida, so if Ida goes 6 miles, then R.U. will go $5 \times 3 = 15$ miles. The ratio of R.U.'s distance to Ida's distance is $15 : 6$, so $\frac{15}{21} = \frac{5}{7}$ of the total distance is walked by R.U. We are given that the total distance they walk is 7 miles, so

$$\text{distance R.U. walks } = \frac{5}{7} \times 7 = \boxed{5} \text{ miles.}$$

15 A triangle has corners $A(0,0)$, $B(12,0)$, and $C(0,6)$. If the midpoint of AB is connected to C and the midpoint of AC is connected to B, they intersect at a point with coordinates (x,y). What is x?

Your Answer	[no answer given]
Correct Answer	4

Explanation

We can use slope-intercept form to find the equations of the two lines, and solve the system of equations to find their intersection. The midpoint of AB is $(6,0)$, and so the line through that point and C has slope -1, and hence equation $y=-x+6$.

The midpoint of AC is $(0,3)$, and so the line through that point and B has slope $-\frac{1}{4}$, and hence equation $y=-\frac{1}{4}x+3$.

Solving this system of equations, we have

$$\begin{aligned} -x+6 &= -\frac{1}{4}x+3 \\ 3 &= \frac{3}{4}x \\ x &= \boxed{4}. \end{aligned}$$

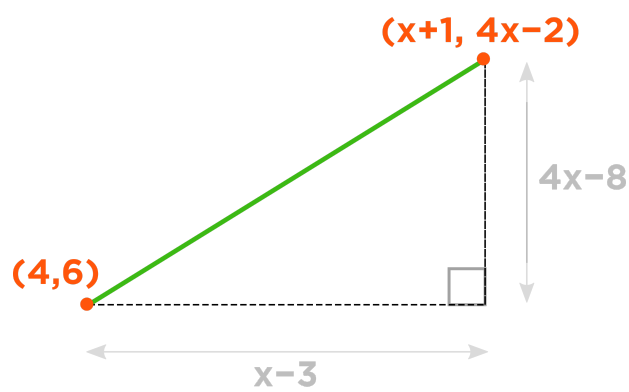
Math note. It turns out that whenever we intersect the *medians* of a triangle, the intersection point along each median is $\frac{1}{3}$ of the way from each side to the opposite vertex of the triangle.

16 The length of the straight line between $(4, 6)$ and $(x+1, 4x-2)$ is $\sqrt{151x}$.

What is the sum of the possible values of x?

✕	**Your Answer**	[no answer given]
	Correct Answer	13

Explanation



Pythagorean Theorem

$(x - 3)^2 + (4x - 8)^2 = \text{hypotenuse}^2$

The distance between $(4, 6)$ and $(x+1, 4x-2)$ is the hypotenuse of the right triangle with one leg equal to the y difference between the points, and the other leg equal to the x difference between the points.

$$\begin{aligned}\text{distance} &= \sqrt{(4x-2-6)^2 + (x+1-4)^2} \\ &= \sqrt{(4x-8)^2 + (x-3)^2} \\ &= \sqrt{16x^2 - 64x + 64 + x^2 - 6x + 9} \\ &= \sqrt{17x^2 - 70x + 73}\end{aligned}$$

We are given that this distance equals $\sqrt{151x}$, so

$$\text{distance} = \sqrt{17x^2 - 70x + 73} = \sqrt{151x}$$

Squaring both sides and combining the x terms, we get

$$17x^2 - 221x + 73 = 0.$$

We want the sum of the values of x, so let's put this into the standard form by dividing the equation by 17.

$$x^2 - 13x + \frac{73}{17} = 0$$

By Viete's Formulas, the coefficient of the x term is equal to the negative sum of the roots of the quadratic, so the sum of the roots is equal to $\boxed{13}$.

17 A straight line passes through the points $(-10, \frac{1}{3})$ and $(-2, \frac{17}{3})$. What is the x-intercept of another line perpendicular to this first line which passes through the midpoint between $(-10, \frac{1}{3})$ and $(-2, \frac{17}{3})$?

Your Answer [no answer given]

Correct Answer -4

Explanation

The midpoint between $(-10, \frac{1}{3})$ and $(-2, \frac{17}{3})$ is $(\frac{-10-2}{2}, \frac{\frac{1}{3}+\frac{17}{3}}{2}) = (-6, 3)$. The slope of the line that passes through these two points is

$$\frac{\frac{17}{3} - \frac{1}{3}}{-2-(-10)} = \frac{16}{3} \times \frac{1}{8} = \frac{2}{3}.$$

So the other line, being perpendicular to this one, has slope of $-\frac{3}{2}$. Since it passes through the midpoint, which is $(-6, 3)$, we can solve the equation

$$y = -\frac{3}{2}x + c,$$

$$3 = -\frac{3}{2}(-6) + c$$

$$3 = 9 + c \implies c = -6.$$

The x-intercept is the value of the function when $y = 0$, so the x-intercept is equal to

$$0 = -\frac{3}{2}x - 6 \implies x = -\frac{12}{3} = \boxed{-4}.$$

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 Find the smallest positive integer n such that $n^2 - 20n + 64$ is a positive square.

Your Answer	[no answer given]
Correct Answer	20

Explanation

Solution 1: (Quadratic Formula)

Using the fact that $n^2 - 20n + 64$ is a square, we can write

$$n^2 - 20n + 64 = x^2,$$

where x is an integer.

This looks like a quadratic in variable n, since we have an n^2 term. Let's put everything to one side and use the Quadratic Formula:

$$n^2 - 20n + 64 - x^2 = 0.$$

The quadratic formula tells us that the solutions of this equation are

$$\begin{aligned} n &= \frac{20 \pm \sqrt{(-20)^2 - 4\left(64 - x^2\right)}}{2} \\ &= \frac{20 \pm \sqrt{400 - 256 + 4x^2}}{2} \\ &= \frac{20 \pm \sqrt{144 + 4x^2}}{2} \end{aligned}$$

Divide out the 2 to simplify it like so:

$$n = 10 \pm \sqrt{36 + x^2}$$

We know n must be an integer, so $\sqrt{36 + x^2}$ must be an integer, which implies $36 + x^2$ is a square. You can now test the possibilities for integer x :

$$
\begin{aligned}
x = 1 &\implies \sqrt{36 + 1^2} = \sqrt{37} \\
x = 2 &\implies \sqrt{36 + 2^2} = \sqrt{40} \\
x = 3 &\implies \sqrt{36 + 3^2} = \sqrt{45} \\
x = 4 &\implies \sqrt{36 + 4^2} = \sqrt{52} \\
x = 5 &\implies \sqrt{36 + 5^2} = \sqrt{61} \\
x = 6 &\implies \sqrt{36 + 6^2} = \sqrt{72} \\
x = 7 &\implies \sqrt{36 + 7^2} = \sqrt{85} \\
x = 8 &\implies \sqrt{36 + 8^2} = \boxed{\sqrt{100}}
\end{aligned}
$$

Thus $x = 8$ works, and plugging this back into $n = 10 \pm \sqrt{36 + x^2}$, we get

$$
\begin{aligned}
n &= 10 \pm \sqrt{36 + 8^2} \\
&= 10 \pm \sqrt{36 + 64} \\
&= 10 \pm \sqrt{100} \\
&= 10 \pm 10 \\
&= 0 \text{ or } \boxed{20}
\end{aligned}
$$

It doesn't make sense for n to be zero, so $n = \boxed{20}$.

Solution 2: (Factoring)

You may prefer to factor $n^2 - 20n + 64$ as

$$(n - 4)(n - 16)$$

We want this to be a positive square, and n must be positive. Looking at this, if n is between 4 and 16, inclusive (meaning including 4 and 16), then $(n - 4)(n - 16)$ will not be positive. We don't want $n = 4$ or $n = 16$, since then n will be zero, and n is supposed to be positive. We only have to consider values of $1 \leq n \leq 3$ and $n > 16$.

$$
\begin{aligned}
n = 1 &\implies (1 - 4)(1 - 16) = -3 \times -15 = 45 \\
n = 2 &\implies (2 - 4)(2 - 16) = -2 \times -14 = 28 \\
n = 3 &\implies (3 - 4)(3 - 16) = -1 \times -13 = 13
\end{aligned}
$$

None of these are squares, so we can now try $n > 16$.

$$
\begin{aligned}
n = 17 &\implies (17 - 4)(17 - 16) = 13 \times 1 = 13 \\
n = 18 &\implies (18 - 4)(18 - 16) = 14 \times 2 = 28 \\
n = 19 &\implies (19 - 4)(19 - 16) = 15 \times 3 = 45 \\
n = 20 &\implies (20 - 4)(20 - 16) = 16 \times 4 = \boxed{64}
\end{aligned}
$$

It's interesting to note that we are trying values of n directly, unlike in Solution 1, where we were trying values of x. We get a square when $n = 20$, so we know our answer is $\boxed{20}$.

19 What is the value of b that makes

$$42ab + 24b = 35a + 20$$

true for all values of a?

Your answer should be a fraction in simplest terms; e.g. if the answer is $\frac{1}{2}$, you would put 1 in the first blank and 2 in the second blank.

_1._ / _2._

Your Answer

1. [no answer given]
2. [no answer given]

Correct Answer

1. 5
2. 6

Explanation

None.

20 Suppose $xy = 9$ and $x + \frac{1}{y} = 100$. Find the value of $y + \frac{1}{x}$.
Your answer should be a fraction in simplest terms, e.g. if the answer were 1/2, you would put 1 in the first blank, and 2 in the second blank.

1. / 2.

Your Answer

1. [no answer given]
2. [no answer given]

Correct Answer

1. 1
2. 9

Explanation

$$\begin{aligned}\left(x+\frac{1}{y}\right)\left(y+\frac{1}{x}\right) &= xy + x\left(\frac{1}{x}\right) + \left(\frac{1}{y}\right)y + \left(\frac{1}{y}\right)\left(\frac{1}{y}\right)\\ &= xy + 1 + 1 + \frac{1}{xy}\\ &= xy + 2 + \frac{1}{xy}\end{aligned}$$

$$\begin{aligned}\left(x+\frac{1}{y}\right)\left(y+\frac{1}{x}\right) &= 9 + 2 + \frac{1}{9}\\ &= 11 + \frac{1}{9} + \frac{100}{9}\\ 100 \times \left(y+\frac{1}{x}\right) &= \frac{100}{9}\\ \implies \left(y+\frac{1}{x}\right) &= \boxed{\frac{1}{9}}\end{aligned}$$

Summary

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